

ORIGINAL RESEARCH

The Role of Coronary Artery Calcium in Statin Eligibility Based on the American Heart Association's PREVENT Calculator

Insights From MESA

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ABSTRACT

BACKGROUND The 2026 American College of Cardiology/American Heart Association/multisociety dyslipidemia guideline incorporates the PREVENT atherosclerotic cardiovascular disease (ASCVD) equations for estimation of 10-year ASCVD risk, replacing the Pooled Cohort Equations used in previous prevention guidelines. Coronary artery calcium (CAC) has historically been used to refine risk assessment when treatment decisions are uncertain.

OBJECTIVES In this study, the authors sought to evaluate cardiovascular event rates across CAC strata within statin eligibility groups and 10-year PREVENT risk categories.

METHODS Individuals from the MESA (Multi-Ethnic Study of Atherosclerosis) were included. Three groups were created based on statin eligibility: statin recommended, considered, and not recommended. In addition, participants with a low-density lipoprotein cholesterol 70 to 189 mg/dL without diabetes were grouped by 10-year PREVENT-ASCVD risk categories: <3% (low risk), 3% to <5% (borderline risk), 5% to <10% (intermediate risk), and ≥10% (high risk). Incidence rates per 1,000 person-years were calculated across CAC strata within statin eligibility groups and PREVENT-ASCVD risk categories.

RESULTS Among the 5,698 participants included in this analysis, the mean age was 61.5 ± 10.3 years and 52.8% were female. Statin therapy was not recommended in 1,924 participants (33.8%), considered in 897 participants (15.7%), and recommended in 2,877 participants (50.5%). The event rates per 1,000 person-years in those without CAC in the not recommended, considered, and recommended statin groups were 1.2, 2.7, and 5.8, respectively. In those with CAC >0, the event rates were 4.5, 5.2, and 16.6, respectively. The event rates per 1,000 person-years in participants with CAC 0 vs CAC >0 were, respectively, 1.1 and 3.0 for 10-year PREVENT-ASCVD risk <3%, 2.7 and 5.2 for 10-year PREVENT-ASCVD risk 3% to <5%, 5.9 and 11.3 for 10-year PREVENT-ASCVD risk 5% to <10%, and 6.2 and 20.9 for 10-year PREVENT-ASCVD risk ≥10%.

CONCLUSIONS In this large multiethnic cohort, CAC provided its greatest clinical value when statin treatment decisions were uncertain, particularly among individuals with borderline predicted risk. Among participants who already met guideline-based treatment thresholds, observed ASCVD event rates remained elevated even when CAC was absent, suggesting that CAC 0 should not generally be used to withhold statin therapy in these individuals. These findings support a guideline-centered role for CAC in the PREVENT era: refining risk near treatment thresholds, contextualizing absolute risk, and informing the intensity of preventive efforts. (JACC. 2026;88:739-748) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS AND ACRONYMS

ASCVD = atherosclerotic
cardiovascular disease

CAC = coronary artery calcium

CT = computed tomography

eGFR = estimated glomerular
filtration rate

HDL-C = high-density
lipoprotein cholesterol

LDL-C = low-density
lipoprotein cholesterol

SBP = systolic blood pressure

The recently released 2026 American College of Cardiology (ACC)/American Heart Association(AHA)/multi-society guideline on the management of dyslipidemia incorporates the AHA Predicting Risk of Cardiovascular Disease Events (PREVENT) equations for estimation of 10-year atherosclerotic cardiovascular disease (ASCVD) risk.^{1,2} This represents an important change in primary-prevention risk assessment and in the framework used to guide statin therapy.^{1,2} Within the guideline, statin therapy is considered in individuals with borderline predicted risk and recommended in those with intermediate or high predicted risk.¹ When uncertainty remains regarding statin initiation in borderline or intermediate risk individuals, coronary artery calcium (CAC) testing may be used for further risk stratification.¹

SEE PAGE 749

Previous studies have shown that CAC meaningfully refines ASCVD risk assessment, particularly when treatment decisions are uncertain.³ However, most of these data were generated in the era of the Pooled Cohort Equations.³ Accordingly, the role of CAC within PREVENT-based treatment frameworks remains incompletely defined. To address this gap, we used the MESA (Multi-Ethnic Study of Atherosclerosis), a large community-based cohort with detailed phenotyping and long-term follow-up, to evaluate how CAC relates to observed ASCVD event rates and atherosclerotic burden across 10-year PREVENT-ASCVD risk categories and statin eligibility groups defined by the 2026 ACC/AHA/multisociety dyslipidemia guideline.

METHODS

STUDY POPULATION. The methods and study design from MESA have been previously described.⁴ MESA is a community-based cohort that enrolled 6,814 men and women from 6 United States communities from 2000 to 2002.⁴ Participants were 45 to 84 years of age and free of clinical ASCVD at the time of enrollment.⁴ The MESA protocol was approved by the Institutional Review Boards at each of the participating field centers. Informed consent was obtained from every participant. For this analysis, we included 5,698 participants with complete baseline data necessary to calculate the 10-year PREVENT-ASCVD risk scores: age, sex, total cholesterol, high-density lipoprotein cholesterol (HDL-C), systolic blood pressure (SBP), estimated glomerular filtration rate (eGFR), diabetes, current smoking, and antihypertensive medication, as well as low-density lipoprotein cholesterol (LDL-C) for determination of statin eligibility. A total of 106 participants were excluded owing to missing statin recommendation. Among those, 56 were excluded owing to missing 10-year PREVENT-ASCVD risk and 50 owing to missing LDL-C. There were no participants with missing baseline CAC. This study was a complete case analysis.

The PREVENT-based risk categories were defined per the 2026 ACC/AHA/multisociety guideline on the management of dyslipidemia as follows: <3% (low risk), 3% to <5% (borderline risk), 5% to <10% (intermediate risk), ≥10% (high risk).¹ The guideline identifies a 10-year PREVENT-ASCVD risk threshold of ≥3% as the level at which statin therapy should be considered.¹ Using these thresholds and other guideline-based criteria, we created the following statin eligibility groups: statin recommended, statin considered, and statin not recommended.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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- 1) Statin recommended: LDL-C ≥ 190 mg/dL, diabetes, or LDL-C 70–189 mg/dL without diabetes and a 10-year PREVENT-ASCVD risk $\geq 5\%$.
- 2) Statin considered: LDL-C 70–189 mg/dL without diabetes and 10-year PREVENT-ASCVD risk 3% to $<5\%$.
- 3) Statin not recommended: LDL-C 70–189 mg/dL without diabetes and 10-year PREVENT-ASCVD risk $<3\%$ or LDL-C <70 mg/dL without diabetes.¹

MEDICAL HISTORY AND DEMOGRAPHIC DATA.

Medical history and demographic data from participants were obtained from standard questionnaires.⁴ Current smoking was defined as any cigarette use in the past 30 days. All participants had 3 blood pressure measurements using an automated oscillometric sphygmomanometer (Dinamap model Pro 100; Critikon), and the average of the latter 2 measurements was reported.⁴ Diabetes was defined by a fasting plasma glucose ≥ 126 mg/dL or medication use for diabetes treatment.⁴

LABORATORY MEASUREMENTS. Fasting blood samples were obtained from all participants and stored in EDTA-anticoagulant tubes at -70°C (University of Vermont and University of Minnesota).⁴ Total cholesterol and HDL-C were measured with the use of the cholesterol oxidase method (Roche Diagnostic).⁴ LDL-C in participants with triglycerides <400 mg/dL was calculated with the use of the Friedewald equation.⁵ Serum creatinine was measured with the Johnson & Johnson Vitros 950 analyzer (Johnson & Johnson Clinical Diagnostics), and eGFR was calculated according to the Modification of Diet in Renal Disease equation.⁴

CAC SCORING. All participants had baseline CAC testing using either cardiac-gated electron-beam (Northwestern University, University of California at Los Angeles, and Columbia University) or multi-detector electrocardiogram-triggered (Johns Hopkins University, Wake Forest University, and University of Minnesota) computed tomography (CT).⁴ Cardiologists and radiologists interpreted the CT scans at a centralized location (Los Angeles Biomedical Research Institute, Torrance, California, USA).⁴ CTs were phantom calibrated, and calcific lesions following a coronary artery distribution with at least 130 Hounsfield units (HU) across 3 contiguous pixels were included.⁴ These values were then multiplied by a density factor and totaled for a final CAC score.⁴

INCIDENT ASCVD EVENTS. Outcome data were obtained from clinic visits, medical records, obituaries, and participant call-ins.⁴ In addition, trained staff contacted participants via telephone every 9 to

12 months to obtain data on hospital admissions, new diagnoses, cardiovascular procedures, and deaths.⁴ Self-reported data were verified with medical records.⁴ A team of cardiologists, cardiovascular physician epidemiologists, and neurologists adjudicated endpoints.⁴ Incident ASCVD was a composite of myocardial infarction, resuscitated cardiac arrest, stroke (not including transient ischemic attack), stroke death, and coronary heart disease death.⁴ This analysis used event data through the first 10 years of follow-up.

STATISTICAL ANALYSIS. Baseline characteristics across the statin not recommended, statin considered, and statin recommended groups were compared. Continuous variables with a normal distribution were compared with the use of analysis of variance tests. Skewed variables with a continuous distribution were compared with the use of Kruskal-Wallis tests. Categorical variables were compared with the use of chi-square tests. Time-to-ASCVD among statin eligibility groups across CAC strata (CAC 0, CAC 1–99, and CAC ≥ 100) was assessed with the use of Kaplan-Meier plots. Baseline characteristics were shown for the full analytic cohort used to define statin eligibility. Time-to-event analyses were restricted to participants with complete adjudicated ASCVD follow-up data.

Incident rates per 1,000 person-years for statin eligibility groups by CAC strata were reported for the overall cohort and by sex. The models included main effects for CAC group, statin eligibility group, and their interaction. Standardized score process plots and supremum tests were conducted to assess the proportional hazard assumptions. All the covariate-specific tests indicated that the proportional hazard assumptions were reasonable. Unadjusted HRs and corresponding 95% CIs as well as absolute risk difference based on 10-year cumulative incidence and 95% bootstrap CIs were reported. Two sensitivity analyses were performed. One analysis incorporated non-ASCVD death as a competing risk in the proportional subdistribution hazard models. Another analysis included the participants who were already on statin in the Cox model. We note that the main purpose of the analysis was not to develop a prediction model for ASCVD events. Nevertheless, a supplementary analysis was performed with a Cox model that included both continuous PREVENT risk scores (expressed as $\log[-\log(1 - \text{risk})]$ transformations) and CAC group. Wald chi-square test was used to test the statistical significance of CAC group. Calibration was examined using 10-year cumulative incidence based on the risk categories defined as $<3\%$, 3%

TABLE 1 Baseline Characteristics by Statin Eligibility

	Statin Not Recommended (n = 1,924)	Statin Considered (n = 897)	Statin Recommended (n = 2,877)	Total (N = 5,698)	P Value	On Statin at Baseline (n = 1,010)
Age, y	52.4 ± 6.5	59.0 ± 5.5	68.5 ± 8.3	61.5 ± 10.3	<0.001	66.0 ± 8.7
Female	1,236 (64.2)	458 (51.1)	1,317 (45.8)	3,011 (52.8)	<0.001	537 (53.2)
Race/ethnicity					0.019	
White	752 (39.1)	369 (41.1)	1,025 (35.6)	2,146 (37.7)		436 (43.2)
Chinese	244 (12.7)	107 (11.9)	340 (11.8)	691 (12.1)		103 (10.2)
Black	500 (26.0)	227 (25.3)	841 (29.2)	1,568 (27.5)		291 (28.8)
Hispanic	428 (22.2)	194 (21.6)	671 (23.3)	1,293 (22.7)		180 (17.8)
Blood pressure medication	301 (15.6)	233 (26.0)	1,350 (46.9)	1,884 (33.1)	<0.001	619 (61.3)
SBP, mm Hg	114.3 ± 15.6	122.3 ± 16.9	135.0 ± 22.0	126.0 ± 21.5	<0.001	130.1 ± 21.7
DBP, mm Hg	69.8 ± 9.5	72.4 ± 9.9	73.3 ± 10.7	72.0 ± 10.3	<0.001	71.2 ± 9.7
Diabetes	0 (0.0)	0.0 (0.0)	633 (22.0)	633 (11.1)	<0.001	223 (22.1)
Current smoking	218 (11.3)	152 (16.9)	397 (13.8)	767 (13.5)	<0.001	97 (9.6)
Cholesterol, mg/dL	188.3 ± 34.1	198.9 ± 30.4	200.1 ± 36.0	195.9 ± 35.0	<0.001	181.6 ± 33.8
HDL-C, mg/dL	54.4 (16.5)	50.7 (14.8)	49.0 (13.6)	51.1 (15.0)	<0.001	50.8 (13.6)
LDL-C, mg/dL	111.2 (30.8)	122.3 (26.3)	124.4 (32.1)	119.6 (31.4)	<0.001	103.4 (28.3)
eGFR, mL/min/1.73 m ²	78.7 ± 16.3	76.0 ± 13.4	72.3 ± 17.0	75.1 ± 16.5	<0.001	70.8 ± 15.8
CAC >0	435 (22.6)	374 (41.7)	1,868 (64.9)	2,677 (47.0)	<0.001	682 (67.5)
CAC	0.0 (0.0-0.0)	0.0 (0.0-27.3)	26.6 (0.0-198.8)	0.0 (0.0-67.3)	<0.001	33.1 (0.0-211.2)

Values are mean ± SD, n (%), or median (Q1-Q3).

CAC = coronary artery calcium; DBP = diastolic blood pressure; eGFR = estimated glomerular filtration rate; HDL-C = high density lipoprotein-cholesterol; LDL = low density lipoprotein-cholesterol; SBP = systolic blood pressure.

to <5%, 5% to <10%, and ≥10%. Net reclassification improvement (NRI) was calculated to evaluate the incremental value that the CAC group adds to the PREVENT risk. The point estimates and 95% CIs for event, nonevent, and overall NRIs at 10 years were computed with the ASCVD risk categories defined above. Finally, incidence rates per 1,000 person-years across 10-year PREVENT-ASCVD risk categories by presence and absence of CAC were reported in the subset of participants with an LDL-C 70 to 189 mg/dL and without diabetes. The Cox model included the main effect of PREVENT-ASCVD risk categories, presence and absence of CAC, and the interaction term. *P* values of <0.05 were considered to be statistically significant.

RESULTS

A total of 5,698 participants were included with a mean age of 61.5 ± 10.3 years; 52.8% were female. Based on guideline statin eligibility criteria, statin therapy was not recommended in 1,924 participants (33.8%), considered in 897 (15.7%), and recommended in 2,877 (50.5%). Participants recommended for statins were generally older, more likely to be male, and had a higher prevalence of hypertension. The proportions of White, Chinese, Black, and Hispanic participants in each statin recommendation group were similar to those in the overall study

sample. The statin not recommended group had the lowest percentage of current smokers. Total cholesterol and LDL-C were higher, and HDL-C and eGFR were lower, among participants in the statin considered and statin recommended groups (Table 1).

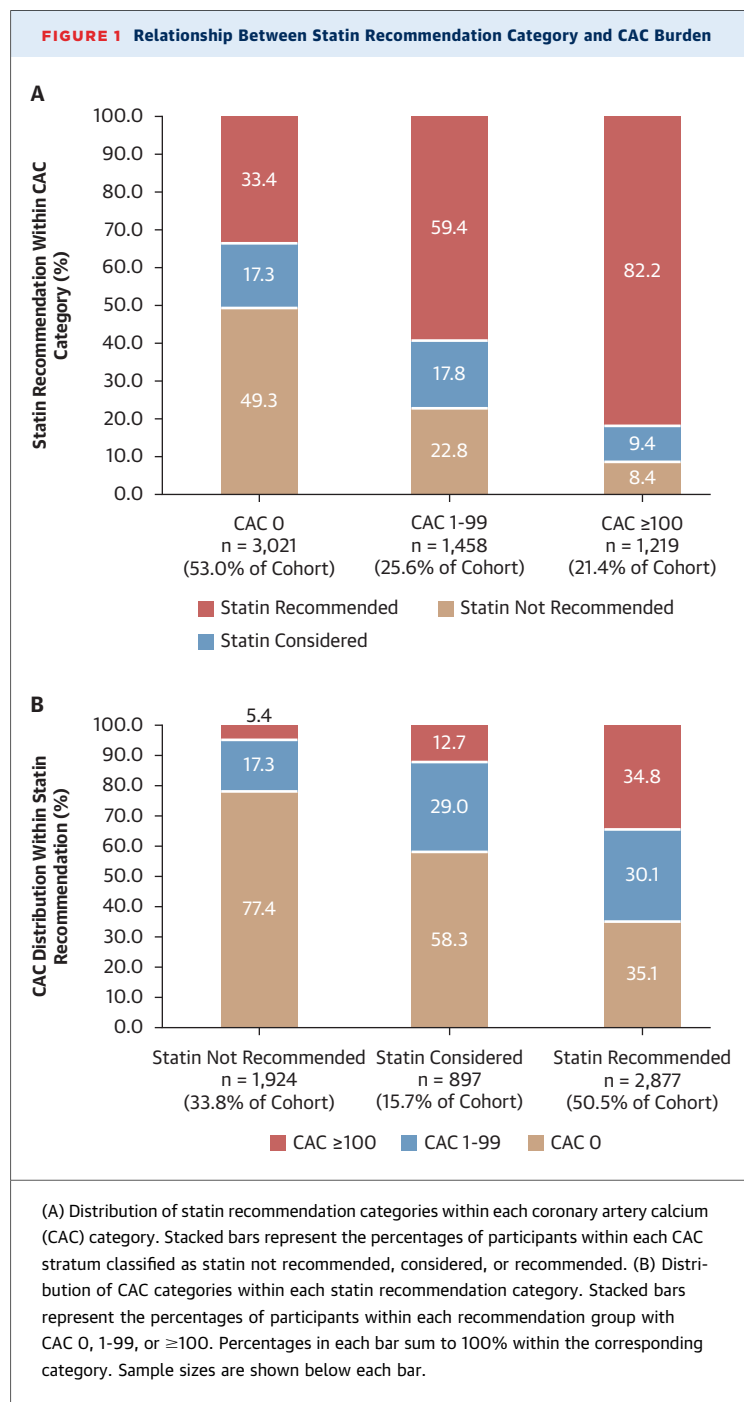
Overall, CAC was present in 2,677 participants (47.0%), with substantial variation across statin eligibility groups. The prevalences of CAC >0 were 22.6% in the statin not recommended group, 41.7% in the statin considered group, and 64.9% in the statin recommended group (Table 1). The distribution of statin recommendation categories across CAC strata and the distribution of CAC categories across statin eligibility groups are shown in Figure 1. Among the 1,924 individuals not recommended statin therapy, the prevalences of CAC 0, CAC 1 to 99, and CAC ≥100 were 77.4%, 17.3%, and 5.4%, respectively (Figure 1). Individuals not recommended statin therapy who had CAC were older, more likely to be male, and had a higher prevalence of hypertension (Supplemental Table 1).

There were 374 incident ASCVD events. Among the 374 incident ASCVD events, 166 (44%) were myocardial infarctions, 149 (40%) strokes, 48 (13%) coronary heart disease deaths, and 11 (3%) resuscitated cardiac arrests. Kaplan-Meier curves demonstrated clear separation in observed ASCVD event rates across CAC categories within each statin eligibility group (Figure 2). The multiplicative interactions between

binary CAC and 3-category CAC with statin eligibility groups were 0.36 and 0.047, respectively. Across all statin eligibility groups, presence of CAC identified individuals with substantially higher observed ASCVD event rates (Figure 3). In the statin not recommended group, the observed event rate was 1.2 per 1,000 person-years in participants with CAC 0 ($n = 1,485$), which was below the threshold for statin consideration, and 4.5 per 1,000 person-years in participants with CAC >0 ($n = 435$), which was above that threshold. Among participants in whom statins were considered, the observed event rate was 2.7 per 1,000 person-years in those with CAC 0 ($n = 520$; again below the threshold for statin consideration, and 5.2 per 1,000 person-years in those with CAC >0 ($n = 373$), above that threshold. Among participants in whom statins were recommended, observed event rates exceeded the threshold for statin consideration regardless of CAC status, at 5.8 per 1,000 person-years for those with CAC 0 ($n = 1,003$) and 16.6 per 1,000 person-years for those with CAC >0 ($n = 1,857$) (Table 2). Similarly, observed event rates exceeded the threshold for statin consideration regardless of CAC status in participants on statins at baseline, at 3.6 per 1,000 person-years for those with CAC 0 ($n = 327$) and 11.2 per 1,000 person-years for those with CAC >0 ($n = 680$) (Supplemental Table 2).

Ten-year ASCVD cumulative incidence and absolute risk differences followed a pattern similar to the event rates across statin eligibility groups (Supplemental Table 3). In a competing-risk sensitivity analysis incorporating non-ASCVD death, results were overall similar (Supplemental Table 4). The pattern of higher observed ASCVD event rates with increasing CAC burden was generally consistent by sex. In the statin considered group, absence of CAC lowered risk below the threshold for statin initiation only in men; however, this analysis was exploratory and limited by low events (Supplemental Table 5).

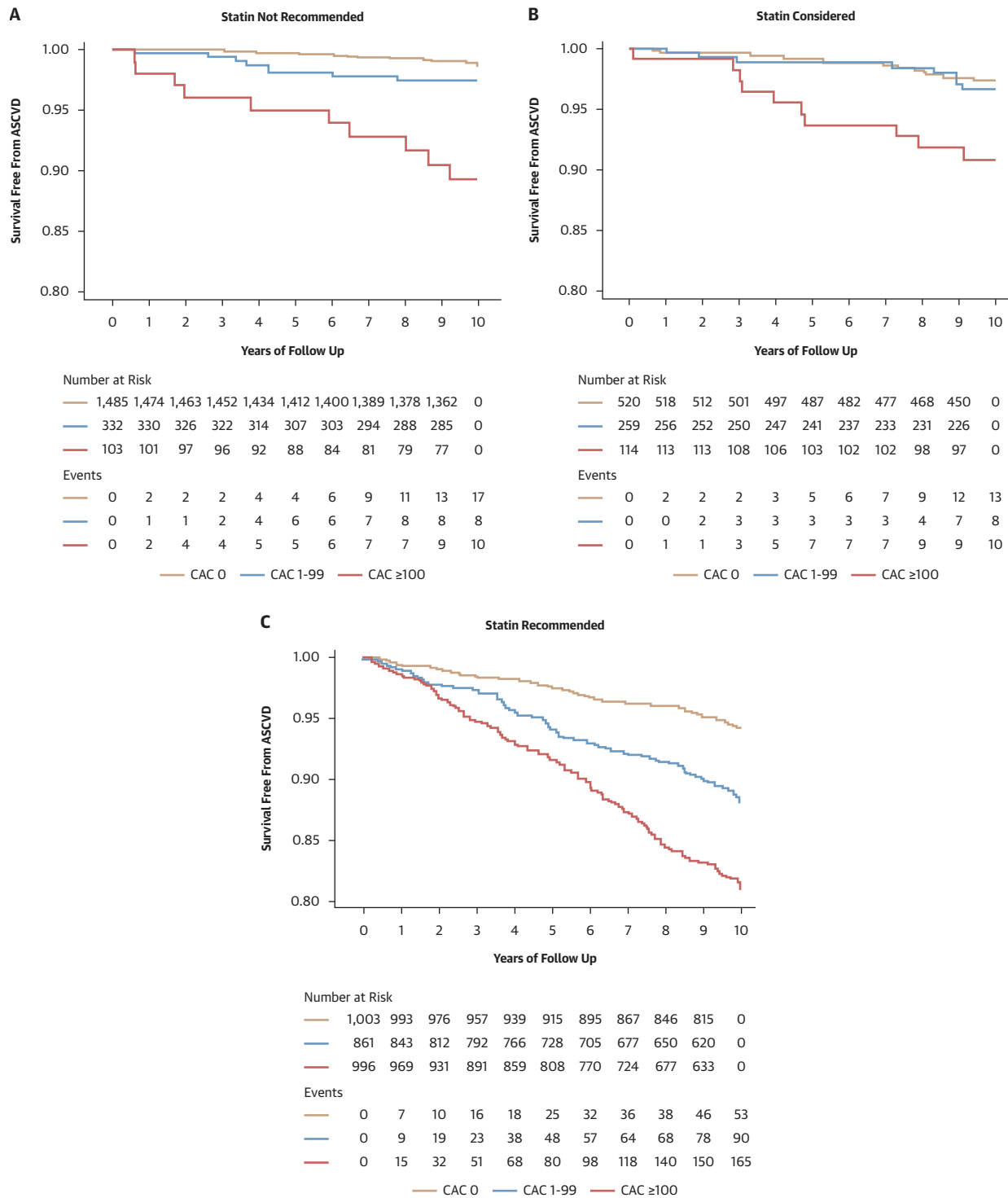
Addition of CAC to PREVENT-based risk categories resulted in a categoric NRI of 0.124 (95% CI: 0.064-0.183) (Supplemental Tables 6 and 7). This improvement was driven by appropriate downward reclassification among individuals who did not experience ASCVD events, with a nonevent NRI of 0.062 (95% CI: 0.046-0.078) and upward reclassification among individuals who did experience ASCVD events, with an event NRI of 0.062 (95% CI: 0.004-0.119). The calibration slope for the 10-year PREVENT-ASCVD + CAC model was 0.81 (95% CI: 0.61-1.01) (Supplemental Table 8). Wald chi-square testing indicated that the CAC group remained significantly associated with



incident ASCVD events after accounting for continuous PREVENT risk ($P < 0.001$).

Observed event rates for 10-year PREVENT-ASCVD risk categories by presence of CAC in participants with an LDL-C of 70-189 mg/dL without diabetes are presented in Table 3. The multiplicative interaction between binary CAC 10-year PREVENT-ASCVD risk categories was 0.47. Among individuals with 10-year

FIGURE 2 Survival Curves Free From ASCVD Based on CAC Strata Across Statin Eligibility Groups



Survival free from ASCVD was lowest in those with CAC ≥ 100 for (A) statin not recommended, (B) statin considered, and (C) statin recommended groups. Number at risk and events are detailed below each survival plot. ASCVD = atherosclerotic cardiovascular disease; CAC = coronary artery calcium.

PREVENT-ASCVD risk of 5% to <10% or $\geq 10\%$, event rates exceeded 3 per 1,000 person-years regardless of CAC. In the PREVENT-ASCVD risk 3% to <5% group, the event rates per 1,000 person-years were 2.7 (95% CI: 1.6-4.6) for CAC 0 and 5.2 (95% CI: 3.3-8.3) for CAC >0. Among individuals with PREVENT-ASCVD risk <3%, event rates were 1.1 (95% CI: 0.6-1.8) for CAC 0 and 3.0 (95% CI: 1.6-5.6) for CAC >0 (**Central Illustration**).

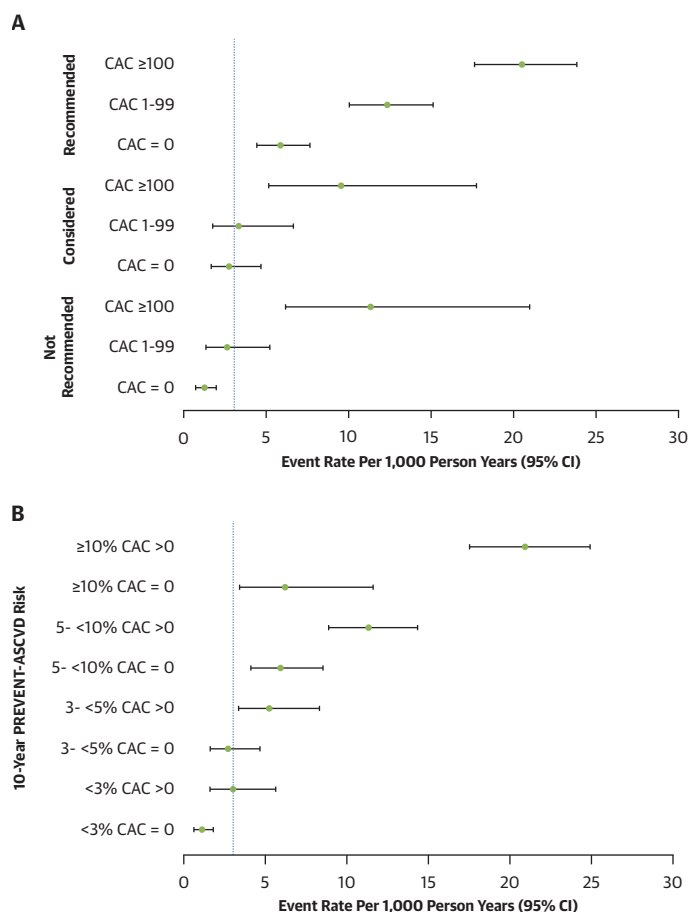
DISCUSSION

In this large multiethnic primary-prevention cohort, there was substantial heterogeneity in observed ASCVD risk across statin eligibility groups. The most clinically important finding was that CAC appeared to be most informative when treatment decisions were uncertain. Among participants in whom statin therapy was considered (borderline 10-year PREVENT-ASCVD risk of 3% to <5%), absence of CAC identified observed 10-year ASCVD event rates below the threshold for statin initiation. In contrast, among participants in whom statin therapy was recommended (10-year PREVENT-ASCVD risk $\geq 5\%$), observed event rates remained above the threshold for statin initiation regardless of CAC status. These findings suggest that CAC may be most useful for refining absolute risk when predicted risk is near a treatment threshold, whereas in higher risk individuals it may be more useful for contextualizing expected benefit and informing the intensity of preventive strategies rather than altering treatment eligibility.

The 2026 dyslipidemia guideline gives CAC an important role in borderline and intermediate predicted risk groups when uncertainty regarding statin initiation remains after clinician-patient discussion.¹ Our findings support that framework. In participants with borderline predicted risk, absence of CAC identified a subgroup with observed event rates below the threshold for statin initiation, supporting the use of CAC to help withhold or defer statin therapy when the decision is uncertain. In contrast, among individuals with intermediate predicted risk, observed event rates were above the treatment threshold whether CAC was absent or present. In this setting, CAC appears less useful for changing the basic treatment recommendation and more useful for contextualizing absolute risk, facilitating shared decision making, and informing the intensity of risk factor modification.

Among individuals with low 10-year predicted PREVENT-ASCVD risk <3%, or among those not recommended statin therapy, observed event rates were low overall. However, the presence of CAC identified

FIGURE 3 Event Rates Among Statin Eligibility and 10-Year PREVENT ASCVD Risk Groups



(A) Event rates per 1,000 person-years among statin eligibility groups, stratified by CAC. (B) Event rates per 1,000 person-years among 10-year PREVENT-ASCVD risk groups, stratified by CAC. The threshold for statin consideration (blue dotted line) is at an ASCVD event rate of 3 per 1,000 person-years. Abbreviations as in **Figure 2**.

a small subgroup with higher observed event rates than suggested by estimated risk alone. Importantly, this was not true uniformly across all CAC-positive individuals in this group. Participants not recommended statin therapy who had CAC 1 to 99 had higher event rates than those with CAC 0, but their observed risk remained below the threshold for statin consideration. In contrast, the smaller subgroup with CAC ≥ 100 had substantially higher observed event rates, exceeding the threshold for statin initiation. These findings should be interpreted cautiously because this subgroup was small, but they suggest that CAC may identify individuals with more advanced subclinical atherosclerosis whose risk is not fully captured by 10-year risk estimation alone.

TABLE 2 ASCVD Events by CAC Across Statin Eligibility Groups

	N	n	Events (%)	Event Rate per 1,000 Person-Years (95% CI)	Unadjusted HR (95% CI)
Statin not recommended	1,920				
CAC 0		1,485	17 (1.1)	1.2 (0.7-1.9)	Reference
CAC 1-99		332	8 (2.4)	2.6 (1.3-5.2)	2.2 (0.9-5.0)
CAC ≥100		103	10 (9.7)	11.3 (6.1-21.0)	9.5 (4.4-20.8)
CAC >0		435	18 (4.1)	4.5 (2.9-7.2)	3.8 (2.0-7.4)
Statin considered	893				
CAC 0		520	13 (2.5)	2.7 (1.6-4.6)	Reference
CAC 1-99		259	8 (3.1)	3.3 (1.7-6.6)	1.2 (0.5-3.0)
CAC ≥100		114	10 (8.8)	9.5 (5.1-17.7)	3.6 (1.6-8.2)
CAC >0		373	18 (4.8)	5.2 (3.3-8.3)	2.0 (1.0-4.0)
Statin recommended	2,799				
CAC 0		1,003	53 (5.3)	5.8 (4.4-7.6)	Reference
CAC 1-99		861	90 (10.5)	12.3 (10.0-15.1)	2.1 (1.5-3.0)
CAC ≥100		996	165 (16.6)	20.5 (17.6-23.8)	3.6 (2.6-4.8)
CAC >0		1,857	255 (13.7)	16.6 (14.7-18.7)	2.9 (2.1-3.9)

Event rates per 1,000 person-years across CAC categories within statin not recommended, statin considered, and statin recommended groups. Unadjusted Cox proportional hazards ratios are displayed with CAC 0 as the reference group. The slightly smaller sample size reflects participants without complete event follow-up.

ASCVD = atherosclerotic cardiovascular disease; CAC = coronary artery calcium.

In such individuals, longer-term risk assessment may be particularly relevant. The 2026 dyslipidemia guideline gives a class 2a recommendation to consider statin therapy in selected individuals with low 10-year PREVENT-ASCVD risk if LDL-C is 160 to 189 mg/dL or 30-year PREVENT-ASCVD risk is elevated.¹ Previous work in MESA has shown that individuals with low 10-year PREVENT-ASCVD risk but elevated 30-year PREVENT-ASCVD risk have a higher prevalence of CAC and greater incident ASCVD risk.⁶ Taken together, these observations suggest that when CAC is elevated despite low estimated 10-year risk, longer-horizon risk assessment may help place that finding into a more clinically meaningful framework.

Previous work by Nasir et al in MESA demonstrated a strong relationship between CAC burden and ASCVD risk when risk was estimated using the

Pooled Cohort Equations.³ Our findings extend those observations to the PREVENT era. Although PREVENT represents an advance in population-level risk estimation, direct imaging of coronary atherosclerotic burden continues to provide clinically relevant information that is not fully captured by clinical risk equations alone. As PREVENT-based risk assessment is incorporated into practice, the role of CAC may evolve from primarily serving as a tool to reclassify individuals around treatment thresholds to serving more broadly as a tool to contextualize atherosclerotic burden and absolute event risk within guideline-defined treatment groups. In this way, CAC may help clinicians interpret heterogeneity in observed risk that persists even within contemporary PREVENT-based treatment frameworks.

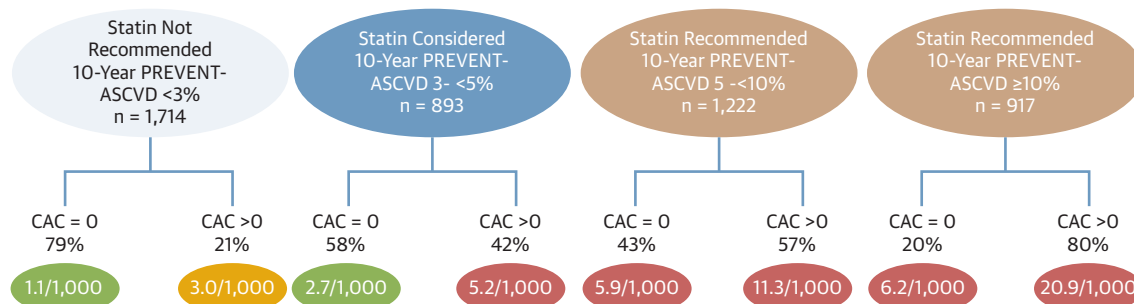
STUDY LIMITATIONS. The present study has several limitations. First, baseline MESA data were used to

TABLE 3 Observed ASCVD Event Rates by CAC Across 10-Year PREVENT-ASCVD Risk Groups in Participants With an LDL-C 70 to 189 mg/dL Without Diabetes

10-Year PREVENT-ASCVD Risk Category	CAC 0			CAC >0		
	n	Events (%)	Event Rate per 1,000 Person-Years (95% CI)	n	Events (%)	Event Rate per 1,000 Person-Years (95% CI)
<3%	1,358	14 (1.0)	1.1 (0.6-1.8)	356	10 (2.8)	3.0 (1.6-5.6)
3% to <5%	520	13 (2.5)	2.7 (1.6-4.6)	373	18 (4.8)	5.2 (3.3-8.3)
5% to <10%	522	29 (5.6)	5.9 (4.1-8.5)	700	70 (10.0)	11.3 (8.9-14.3)
≥10%	184	10 (5.4)	6.2 (3.4-11.6)	733	121 (16.5)	20.9 (17.5-24.9)

Event rates per 1,000 person-years by CAC 0 and CAC >0 for 10-year PREVENT-ASCVD <3%, 3% to <5%, 5% to <10%, and ≥10% groups. Abbreviations as in Table 2.

CENTRAL ILLUSTRATION Impact of CAC Based on 10-Year PREVENT-ASCVD Risk Categories in Participants With LDL-C 70-189 mg/dL Without Diabetes



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Observed cardiovascular event rates (per 1,000 person-years) stratified by 10-year PREVENT-ASCVD risk and CAC status. Top colors reflect guideline-based statin recommendation categories, and the bottom colors reflect observed event rates. Absence of CAC in participants with borderline (10-year PREVENT-ASCVD risk 3% to <5%) lowered observed event rates below the level of statin initiation. ASCVD = atherosclerotic cardiovascular disease; CAC = coronary artery calcium; LDL-C = low-density lipoprotein cholesterol.

calculate 10-year PREVENT-ASCVD risk scores, and those baseline characteristics may not fully reflect current population-level risk factor distributions. Second, CAC results were not blinded from participants or clinicians, which may have influenced subsequent preventive care and attenuated the observed risk associated with CAC. Third, MESA is a relatively healthy primary prevention cohort, which may have contributed to lower absolute ASCVD event rates. Fourth, the ASCVD outcome definition used in this analysis was chosen to align with the outcomes framework used in the derivation and validation of the PREVENT equations. However, the 2026 dyslipidemia guideline uses a broader clinical ASCVD construct; this difference should be considered when extrapolating our findings to broader guideline-defined ASCVD endpoints. Fifth, the statin recommended group included clinically distinct subgroups with different underlying risk structure, and pooling of these subgroups may have contributed to heterogeneity within the statin recommended category. Finally, although statin eligibility groups were based on the 2026 dyslipidemia guideline, our “not recommended,” “considered,” and “recommended” framework represents a simplified operationalization of the guideline Calculate-Personalize-Reclassify and Reassess framework. This classification did not account for all risk-enhancing factors, special populations for whom lipid-lowering therapy may be recommended independently from LDL-C level, including persons living with HIV and those with stage 3 or 4 chronic kidney disease, or other

individualized clinical considerations that inform shared decision making. As a result, some participants may have been misclassified in terms of guideline-based statin eligibility.

CONCLUSIONS

Taken together, these findings support a guideline-centered interpretation of CAC in the PREVENT era. CAC appears to be most informative when treatment decisions remain uncertain, particularly among individuals with borderline predicted risk. In contrast, among participants who already meet PREVENT-based treatment thresholds, CAC 0 should not generally be interpreted as a basis to withhold statin therapy. Rather, CAC may be most useful for contextualizing absolute risk and guiding the intensity of preventive efforts within contemporary prevention frameworks.

DATA AVAILABILITY All data used for this analysis is available on acceptance of approved proposal from MESA: <http://www.mesa-nhlbi.org>.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: In primary-prevention adults evaluated within the 2026 dyslipidemia guideline framework, CAC appears to be most useful when statin decisions remain uncertain, particularly among individuals with borderline 10-year PREVENT-ASCVD risk. In individuals who already meet guideline-based statin treatment thresholds, CAC 0 should not generally be interpreted as a reason to withhold therapy, but CAC burden may help contextualize absolute risk and guide intensity of preventive efforts.

TRANSLATIONAL OUTLOOK 1: Future studies should determine whether CAC-guided decision making within PREVENT-based treatment frameworks improves clinician-patient risk communication, statin initiation, treatment intensification, adherence, and ASCVD outcomes, particularly among patients near guideline-based treatment thresholds.

TRANSLATIONAL OUTLOOK 2: Additional work is needed to define which lower-risk individuals with elevated lifetime or 30-year ASCVD risk derive the greatest clinical value from CAC testing despite low 10-year PREVENT-ASCVD risk.

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KEY WORDS ASCVD, coronary artery calcium, PREVENT, primary prevention, risk calculator, statin eligibility

APPENDIX For supplemental tables, please see the online version of this paper.



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